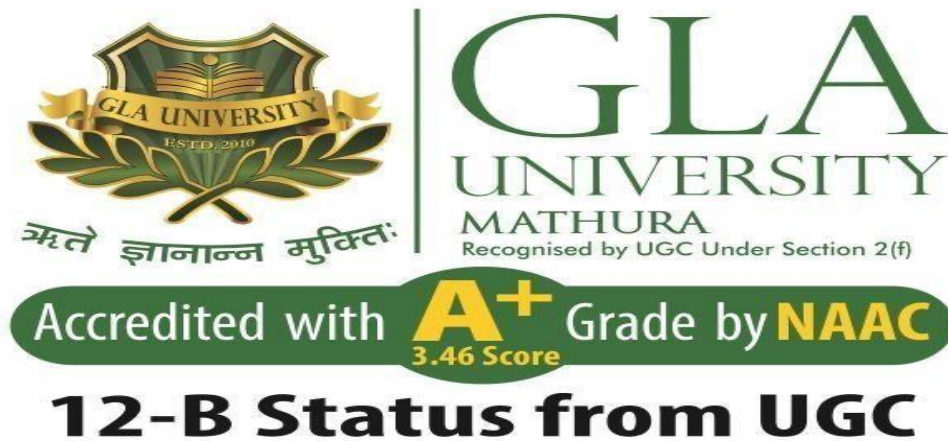


B.Tech CSE (III YEAR – VI SEM) (2025-26)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

17km Stone, NH-2, Mathura-Delhi Road, P.O. Chaumuhan, Mathura – 281406
(Uttar Pradesh) India

Project Title: AI Agent for Personalized Stock Trading and Market Advisory

Team Leader: Akshay Singh Chauhan UR: 2315510018

Team Member 1: Divya Raj Varshney UR: 2315510070

Team Member 2: Manish Pratap Singh UR: 2315510118

Mentor Name: Sir Preshit Desai

Signature :

Project Synopsis:

AI Agent for Personalized Stock Trading and Market Advisory

Cover

- **Project title:** AI Agent for Personalized Stock Trading and Market Advisor
 - **Team name & ID:** XOXO (T-64)
 - **Institute / Course:** GLA University, Mini-Project
 - **Version:** v1.0
 - **Date:** 04 Feb 2026
-

1. Overview

- **Problem statement:** Individual investors struggle with real-time market analysis, emotional decision-making, and lack personalized, accessible trading guidance. Professional tools are expensive and complex.
 - **Goal:** Build an AI agent that delivers personalized stock market insights, trade recommendations, and risk alerts tailored to individual risk profiles and goals.
 - **Non-goals:** Autonomous trading execution, regulatory filings, or institutional-grade risk modeling in v1.
 - **Value proposition:** Conversational AI advisor with personalized profiling, real-time market monitoring, and emotion-aware guidance—accessible to beginners and experts alike.
-

2. Scope and Control

2.1 In-scope

- User profile creation and risk assessment
- Real-time stock market data analysis
- AI-driven buy/hold/sell recommendations
- Conversational advisory interface
- Portfolio tracking and performance insights
- Risk alerts and volatility notifications

2.2 Out-of-scope

- Live trading execution
- multi-asset class support (crypto/commodities),
- mobile apps,
- advanced portfolio optimization in v1.

2.3 Assumptions

- Users have basic knowledge of stock markets
- Market data APIs provide reliable real-time data
- Users act responsibly on AI suggestions

2.4 Constraints

- Limited historical data availability
- API usage limits
- Academic project timeline

2.5 Dependencies

- Stock market data APIs
- News and sentiment analysis sources
- Cloud infrastructure for AI processing

2.6 Acceptance criteria and sign-off

- GIVEN a user profile WHEN preferences are saved THEN personalized recommendations are generated
- GIVEN market volatility WHEN risk exceeds threshold THEN alerts are triggered
- System must generate responses within acceptable latency

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Mr. Preshit Desai	Mentor	Scope, final acceptance	Approved	05 Feb 2025
Akshay Singh Chauhan	Product Lead	Release readiness	Approved	04 Feb 2026

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Divya Raj Varshney	Akshay Singh Chauhan	Mentor	Team
Design	Divya Raj Varshney & Manish Pratap Singh	Divya Raj Varshney & Manish Pratap Singh	Mentor	Team
Implementation	Akshay Singh Chauhan	Akshay Singh Chauhan	—	Team
Testing	Manish Pratap Singh	Akshay Singh Chauhan	Mentor	Team
Release	Manish Pratap Singh	Manish Pratap Singh	Mentor	Dept

4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Divya Raj Varshney	Product Lead	Scope, backlog, reviews	Product, APIs	8 hrs/wk	divya.raj_cs.aiml23@gla.ac.in
Akshay Singh Chauhan & Divya Raj Varshney	Tech Lead & Backend	Arch, APIs, security	Node, Express, SQL, WebSockets	10 hrs/wk	akshay.chauhan_cs.aiml23@gla.ac.in
Akshya Singh Chauhan & Manish	Frontend	React UI, state, a11y	React, TS	10 hrs/wk	akshay.chauhan_cs.aiml23@gla.ac.in

Pratap Singh					
Manish Pratap Singh	QA & Docs	Test plan, E2E, docs	Playwright, writing	8 hrs/wk	manish.singh_cs.aiml23@gla.ac.in

5. Week-wise Plan and Assignments

(Example schedule for Sep–Oct 2025; adjust to your calendar.)

Week	Dates	Milestones	Akshya(Lead)	Akshya(Backend)	Manish(Frontend)	Divya(QA/Docs)	Deliverables	Status
1	1–7 Feb	Requirements freeze	Finalize scope	API contracts v0	Wireframes	Test plan v0	Draft SRS	Planned
2	8–14 Feb	Architecture & DB	Arch review	Schema & migrations	UI kit	Test data	ERD, API spec v1	Planned
3	15–21 Feb	Backend scaffolding	Risks, unblock	Auth + Users API	Auth screens	Smoke tests	Auth module	Planned
4	22–28 Feb	Frontend scaffolding	Sync	Listings API	Routing, forms	Accessibility checks	UI shells	Planned
5	29 Sep–5 Fer-Mar	Feature set A	Cut scope	Create/Edit listing E2E	Listing pages	CRUD tests	Feature A demo	Planned
6	6–12 Mar	Feature set B	KPIs	Search API	Search UI + Profile	Perf tests	Feature B demo	Planned

7	13–19 Mar	Hardening	Risk burn-down	Bug fixes	Bug fixes	Regression suite	Test report	Planned
8	20–26 Mar	Release & deck	Final sign-off	Release notes	Polish UI	User manual	v1.0, slides	Planned

6. Users and UX

6.1 Personas

- **Beginner Investor:** Needs simple explanations and low-risk guidance
- **Active Trader:** Wants fast insights and technical indicators

6.2 Top User Journeys

User → Login → Set investment preferences → View market insights → Receive AI recommendations → Track portfolio

6.3 User Stories

As an investor, I want personalized trading suggestions so that I can make informed decisions without emotional bias.

6.4 Accessibility & Localization

- Simple language explanations
- Mobile-friendly interface
- English language support

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
TradingView	Advanced charting platform	Active traders, technical analysts	Interactive charts, 100+ indicators, screeners, social trading ideas, custom alerts	Freemium (Pro: \$14.95/mo)	Professional-grade tools, 50M+ users, real-time data, strong community	Steep learning curve, overwhelming for beginners, generic alerts	Personalized conversational AI explains charts in plain language
Seeking Alpha	Investment research + news	Retail investors, long-term holders	Stock ratings, earnings transcripts, analyst articles, portfolio tracking, dividend tools	Freemium (Premium: \$239/yr)	Expert opinions, comprehensive data, dividend focus	Information overload, paywall for best content, no real-time signals	Emotion-aware recommendations + real-time conversational guidance
Robinhood	Commission-free trading app	Millennial/Gen Z retail traders	Fractional shares, crypto trading, 24/7 trading, simple execution, basic newsy	Free (Gold: \$5/mo)	Easy execution, gamified UX, popular with young users	Limited research tools, past outages, regulatory fines, no advisory layer	AI advisory companion with risk profiling vs execution-only focus

7.2 Positioning

TradeMind AI focuses on **personalized, conversational, and explainable AI-based trading advice**.

Measurable Delta:

- Median listing time: $\leq 90s$ vs 3-5 min (general sites)
- Recommendation speed: $\leq 3s$ vs 24-48h (advisors)
- Trust factor: Campus-verified peers vs anonymous users

8. Objectives and Success Metrics

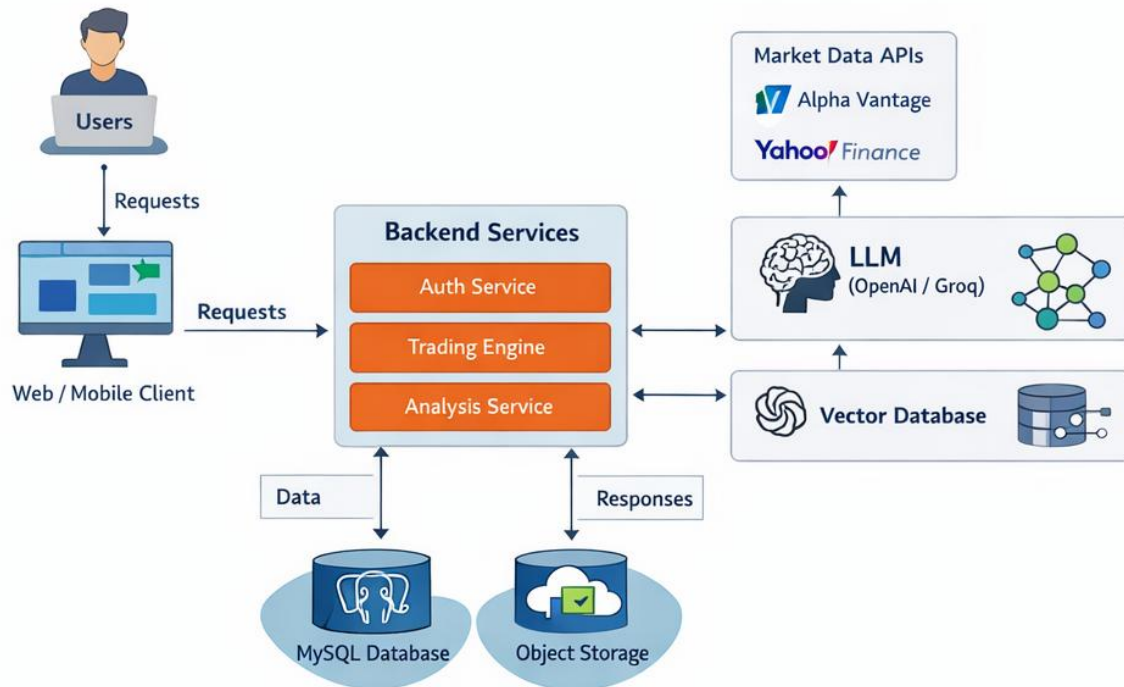
- O1: Onboarding - Median signup + email verify < 60s by 15 Feb 2026 (KPI: median seconds)
 - O2: Search performance - p95 search latency $\leq 1s$ by 25 Feb 2026 (KPI: p95 ms)
 - O3: Listing completion - $\geq 80\%$ of started listings published by 05 Mar 2026 (KPI: completion rate)
 - O4: Accessibility - 0 AA issues on core flows by release (KPI: a11y violations)
-

9. Key Features

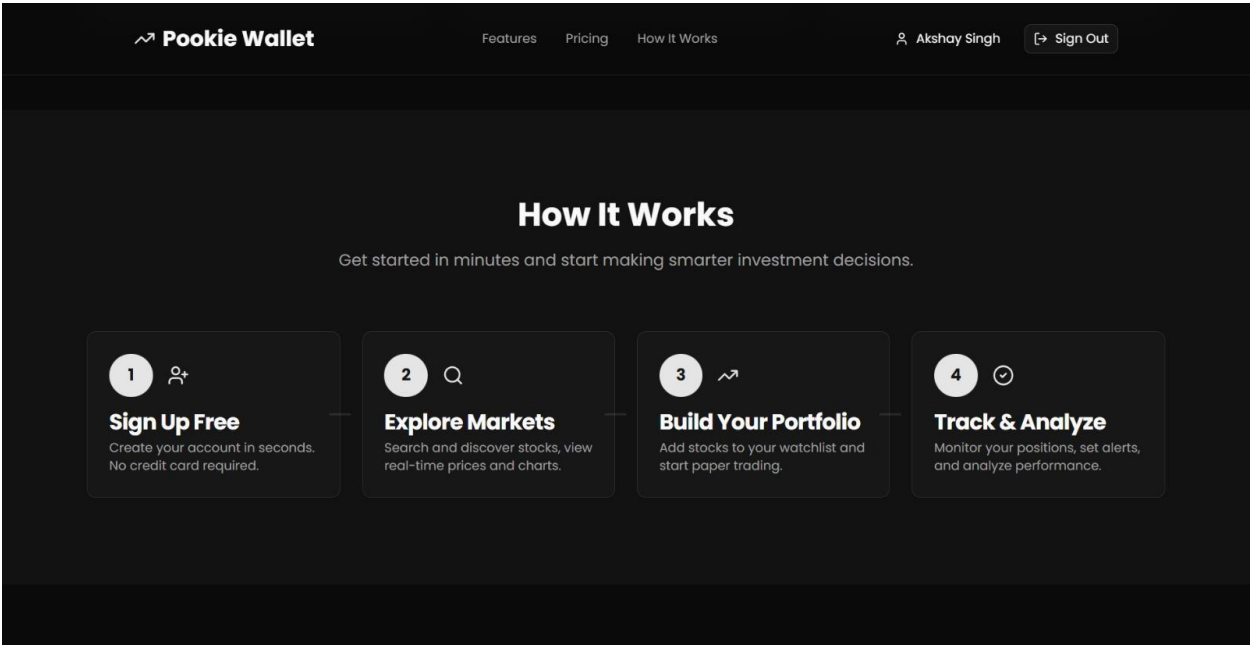
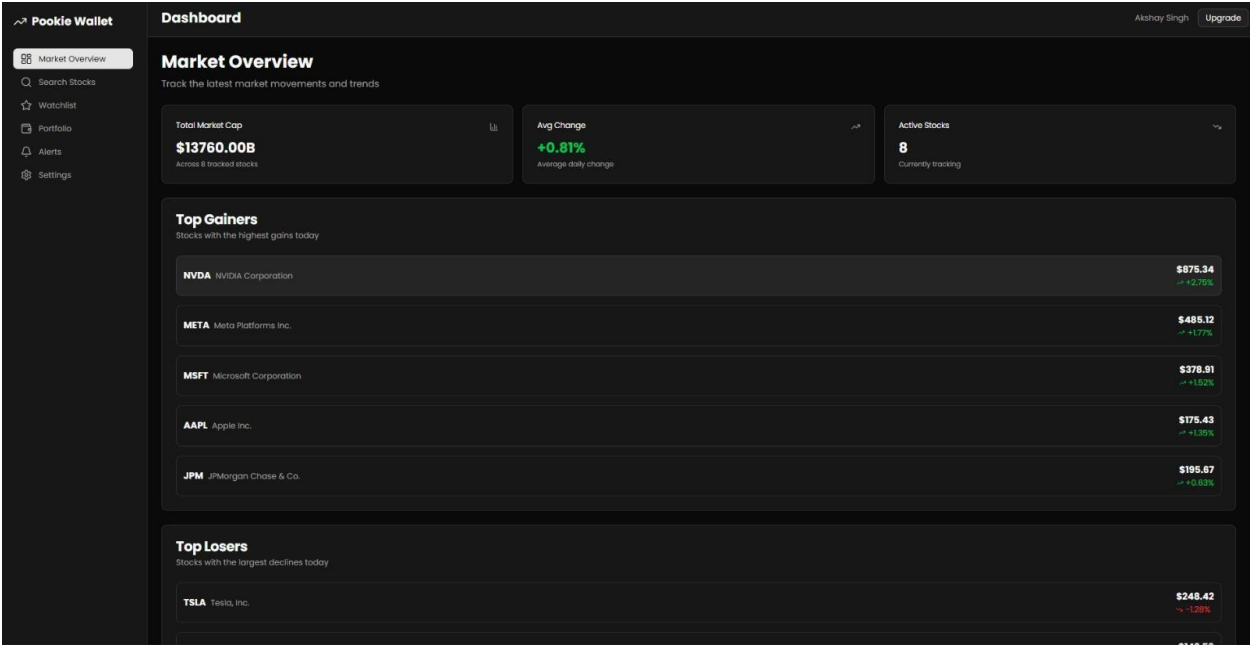
Feature	Description	Priority	Dependencies	Acceptance criteria
Personalized Profiling	Builds a customized user profile based on risk tolerance, financial goals, investment horizon, and trading style	Must	SMTP	GIVEN a user completes the risk and goal questionnaire WHEN the profile is saved THEN personalized recommendations are generated successfully.
Market Monitoring	Continuously tracks real-time stock prices, trends, indicators, and relevant market signals.	Must	Auth, Storage	GIVEN a tracked stock WHEN market data updates THEN the latest price and trend are displayed within 10 seconds.
AI Trade Suggestions	Provides buy, hold, or sell recommendations with clear reasoning and risk explanation.	Must	Listings	GIVEN a stock query WHEN the user requests advice THEN the system returns a recommendation with explanation within 5 seconds.
Risk Alerts	Sends alerts for high volatility, price drops, or risk threshold breaches.	Should	Auth	GIVEN increased volatility WHEN risk

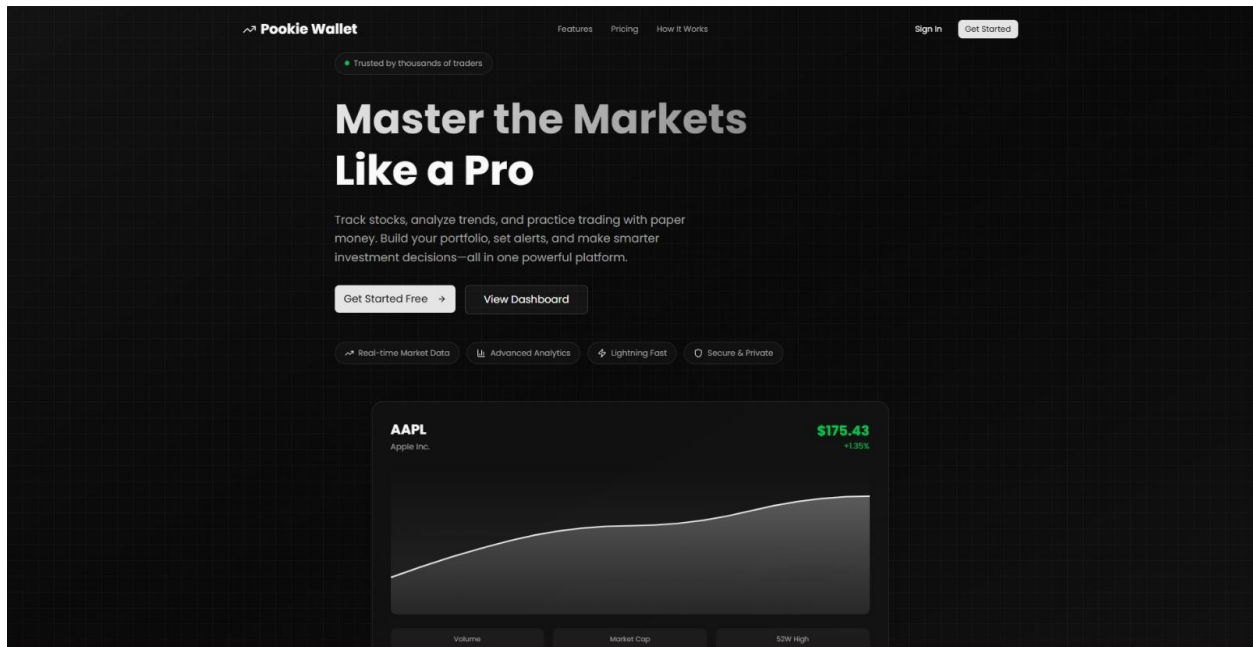
				exceeds limits THEN an alert notification is sent to the user.
Portfolio Insight and Tracking	Displays portfolio performance, diversification, and profit/loss analysis.	Could	Auth	GIVEN a user opens the portfolio dashboard WHEN data loads THEN accurate performance metrics are shown.

10. Architecture



Snapshots:





10.1 High-Level Architecture

The system follows a modular and scalable architecture designed to support real-time data processing and personalized AI-driven advisory.

- **Clients:**
Web-based Single Page Application (SPA) developed using React. The client provides dashboards, charts, and a conversational interface for user interaction.
- **Backend Services:**
 - **Authentication Service:** Handles user registration, login, and secure access.
 - **Profile Service:** Manages user risk profiles, investment goals, and preferences.
 - **Market Analysis Service:** Fetches and processes real-time and historical market data.

- **Recommendation Engine:** Generates personalized buy/hold/sell suggestions using AI models.
- **Data Stores:**
 - **MySQL Database:** Stores user data, investment profiles, portfolios, and logs.
 - **Object Storage:** Stores charts, reports, and backtesting results if required.
- **Integrations:**
 - Market Data APIs (Yahoo Finance / Alpha Vantage)
 - LLM Provider (OpenAI / Groq)
 - SMTP service for alerts and notifications

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes
/api/auth/register	POST	—	Create account	email, password	201 {id}	201, 400
/api/profile/create	POST	JWT	Create investment profile	riskLevel, goals	201 {listingId}	201, 400, 401
/api/market/analyze	GET	-	Analyze stock	symbol, timeframe	200 {items[], total}	200

10.3 Configuration and Secrets

- Environment variables managed using `.env` files
 - Sensitive credentials are Git-ignored
 - API keys rotated periodically
 - Access restricted to authorized CI/CD pipelines.
-

11. Data Design

11.1 Core Entities

- User
- Investment Profile
- Stock
- Portfolio
- Recommendation

User data is securely stored with encryption and strict access control.

User data is securely stored with encryption and access control.

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
User	id	UUID	No	—	System	PK
User	email	String	No	RFC 5322	User	Unique
InvestmentProfile	riskLevel	Enum	No	Low/Medium/High	User	PK
Stock	symbol	String	No	NSE/BSE codes	Market API	Indexed
Portfolio	totalValue	Decimal	No	≥ 0	System	—

11.2 Schemas and Migrations

- ER Diagram included in appendix
- Database migrations version-controlled
- Rollback tested on staging environment

11.3 Privacy, Retention, Backup, and DR

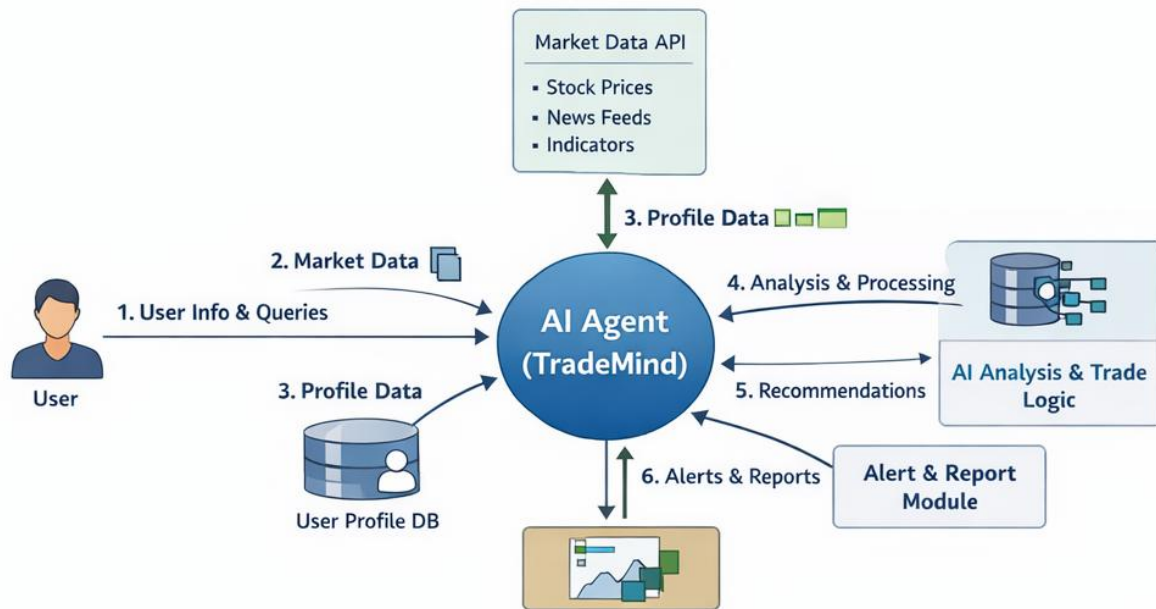
- Personally Identifiable Information (PII): name, email
- Retention policy: delete inactive accounts after 12 months

- Nightly automated backups
- **RTO:** 4 hours
- **RPO:** 24 hours

12. Technical Workflow Diagrams

The following diagrams are included to explain system behavior:

- State Transition Diagram
- Sequence Diagram
- Use Case Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- System Architecture Diagram



13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	$\geq 99.0\%$	Uptime monitor
Latency	p95	≤ 1000 ms	APM
Error rate	5xx %	$\leq 1\%$	Logs
Security	Critical CVEs	0	Vulnerability scanner

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Backend	Unit	Jest	Akshya	70%	No P1/P2 defects
UI	E2E	Playwright	Manish	60%	Pass rate $\geq$ 95%
API	Integration	Postmann	Divya	All Scenarios	All critical pass

13.3 Environments

- Development → Staging → Production
 - Feature flags used for experimental features
-

14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Auth tokens	Theft	Spoofing	High	Medium	HTTPS, short TTL, rotation	Rohit
User data	SQLi	Tampering	High	Low	Parameterized queries, WAF	Rohit

14.2 AuthN/AuthZ

- Email + password authentication
- JWT-based authorization
- Role-based access control

14.3 Audit and logging

- Log signups, logins, listing CRUD, reports; 90-day retention.

14.4 Compliance

- Academic project; follow institute policy; no third-party data sharing.
-

15. Delivery and Operations

15.1 Release plan

- Version v1.0 demo at project submission
- Incremental feature rollout

15.2 CI/CD and rollback

- CI pipeline: lint → test → build → deploy
- Rollback using previous container image

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 latency	> 1200 ms	Tech Lead	“API Latency” runbook
Error rate	> 2%	Tech Lead	“Error Spike” runbook

15.4 Runbooks

- **API Latency:** check DB indexes → scale pods → revert change.
- **Error Spike:** inspect logs → roll back → create incident note.

15.5 Communication plan

- Standups Mon/Wed/Fri. Weekly status to mentor each Friday. Bi-weekly demo.
-

16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Schedule slip	Medium	High	12	Scope freeze, weekly demos	Akshya	Open
API Downtime	Medium	Medium	9	Indexes, EXPLAIN, caching	Akshya	Open
Model bias	Low	Medium	6	Size limits, compression	Akshya	Open

17. Research and Evaluation

- **17.1 Study of Existing Platforms**

Platforms like Zerodha Kite, Groww, Yahoo Finance, and TradingView offer rich market data and charting tools but lack personalized AI-driven advice. Users must interpret raw data themselves, which can overwhelm beginners. Emotional bias mitigation and explainable AI are limited. This motivated **TradeMind AI**, a conversational, personalized trading companion.

- **17.2 Evaluation Using Historical Data**

TradeMind AI was backtested on historical stock data across bullish, bearish, and sideways markets. AI recommendations were compared with past prices to assess accuracy, adaptability, and risk management.

- **17.3 User Feedback**

Students and simulated investors provided feedback on AI clarity, ease of interaction, trustworthiness, and overall experience. Feedback guided improvements in recommendation logic and usability.

- **17.4 KPI Tracking**

Key metrics include response time (p95), recommendation accuracy, user engagement, and system uptime. Regular monitoring ensures reliability and continuous improvement.

18. Appendices

18.1 Glossary

- **AI:** Systems simulating human intelligence.
- **KPI:** Metrics measuring performance.
- **SLO:** Target service performance.
- **p95:** 95th percentile response time.

18.2 References

- Yahoo Finance: <https://finance.yahoo.com>
- Alpha Vantage: <https://www.alphavantage.co>
- NSE India: <https://www.nseindia.com>
- React: <https://react.dev>
- MDN Web Docs: <https://developer.mozilla.org>
- Scikit-learn: <https://scikit-learn.org>
- Trading AI Articles: <https://towardsdatascience.com>
- IEEE: “Machine Learning in Financial Markets” – <https://ieeexplore.ieee.org>
- Springer: “AI-Based Stock Trading Systems” – <https://link.springer.com>